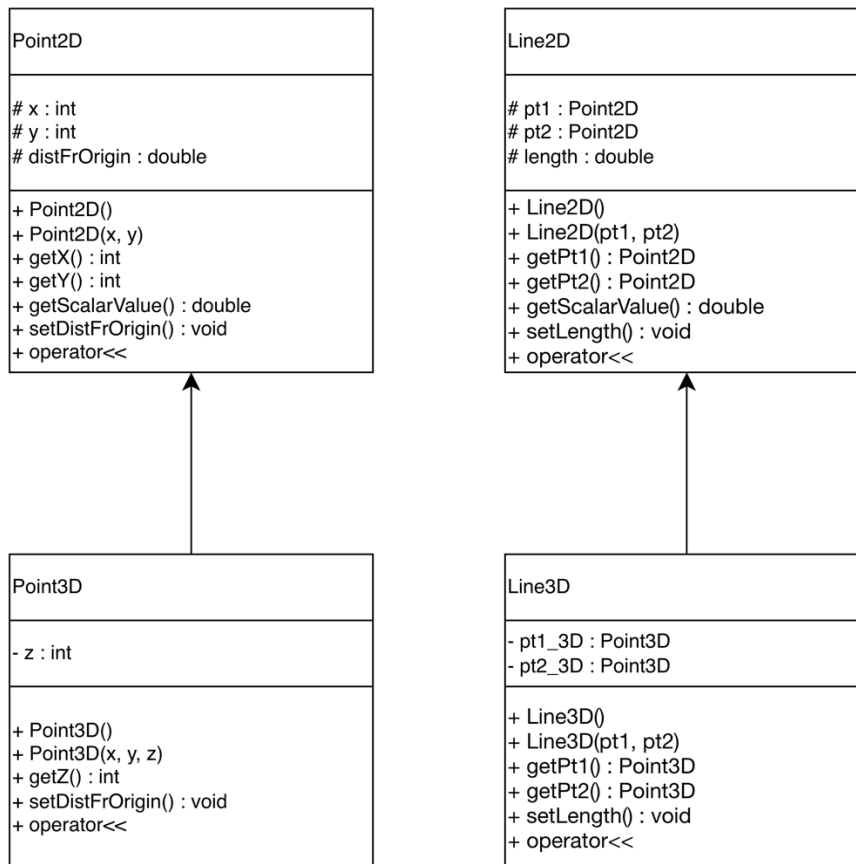


Part 1. Overview Diagram

1. Class Hierarchy

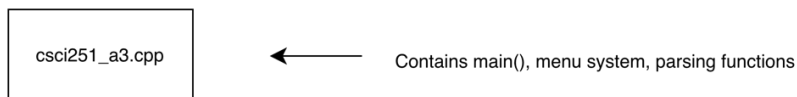


Functions (MyTemplates.h):

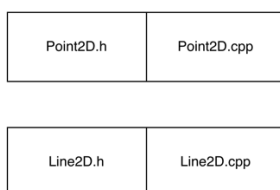
- scalar_difference<T>() - Returns |obj1.getScalarValue() - obj2.getScalarValue()|
- equals<T>() - Specialized for Point2D, Point3D, Line2D, Line3D

2. File Structure

Main Driver:



Base Class:



Derived Classes:

Point3D.h	Point3D.cpp
-----------	-------------

Line3D.h	Line3D.cpp
----------	------------

3. Key Functions in Assn3.cpp

Data Storage (Global Variables):

- `vector<Point2D> point2D_data`
- `vector<Point3D> point3D_data`
- `vector<Line2D> line2D_data`
- `vector<Line3D> line3D_data`

Menu Functions:

- `displayMenu()` - Display 6 menu options with current settings
- `readData()` - Option 1: Read data from file → Parse and remove duplicates
- `specifyFilterCriteria()` - Option 2: Set filter (Point2D/Point3D/Line2D/Line3D)
- `specifySortCriteria()` - Option 3: Set sort attribute (X/Y/Z/Distance/Pt1/Pt2/Length)
- `specifySortOrder()` - Option 4: Set ASC/DSC order
- `viewData()` - Option 5: Display filtered and sorted data
- `storeData()` - Option 6: Save data to file

Parsing Functions:

- `parsePoint2D(line)` - Extract Point2D from string → Handles spaces
- `parsePoint3D(line)` - Extract Point3D from string → Handles spaces
- `parseLine2D(line)` - Extract Line2D from string → Handles spaces
- `parseLine3D(line)` - Extract Line3D from string → Handles spaces

Utility Functions:

- `removeDuplicates()` - Remove all duplicate records using `equals<T>()`

Comparator Functions:

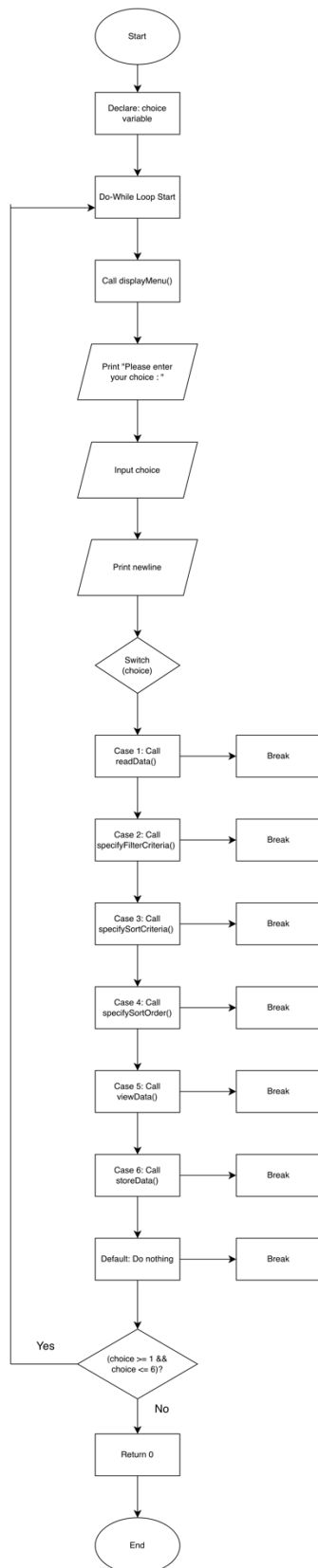
- `comparePoint2D_X()`, `comparePoint2D_Y()`, `comparePoint2D_Dist()`
- `comparePoint3D_X()`, `comparePoint3D_Y()`, `comparePoint3D_Z()`, `comparePoint3D_Dist()`
- `compareLine2D_Pt1XY()`, `compareLine2D_Pt2XY()`, `compareLine2D_Length()`
- `compareLine3D_Pt1XY()`, `compareLine3D_Pt2XY()`, `compareLine3D_Length()`

Main Function:

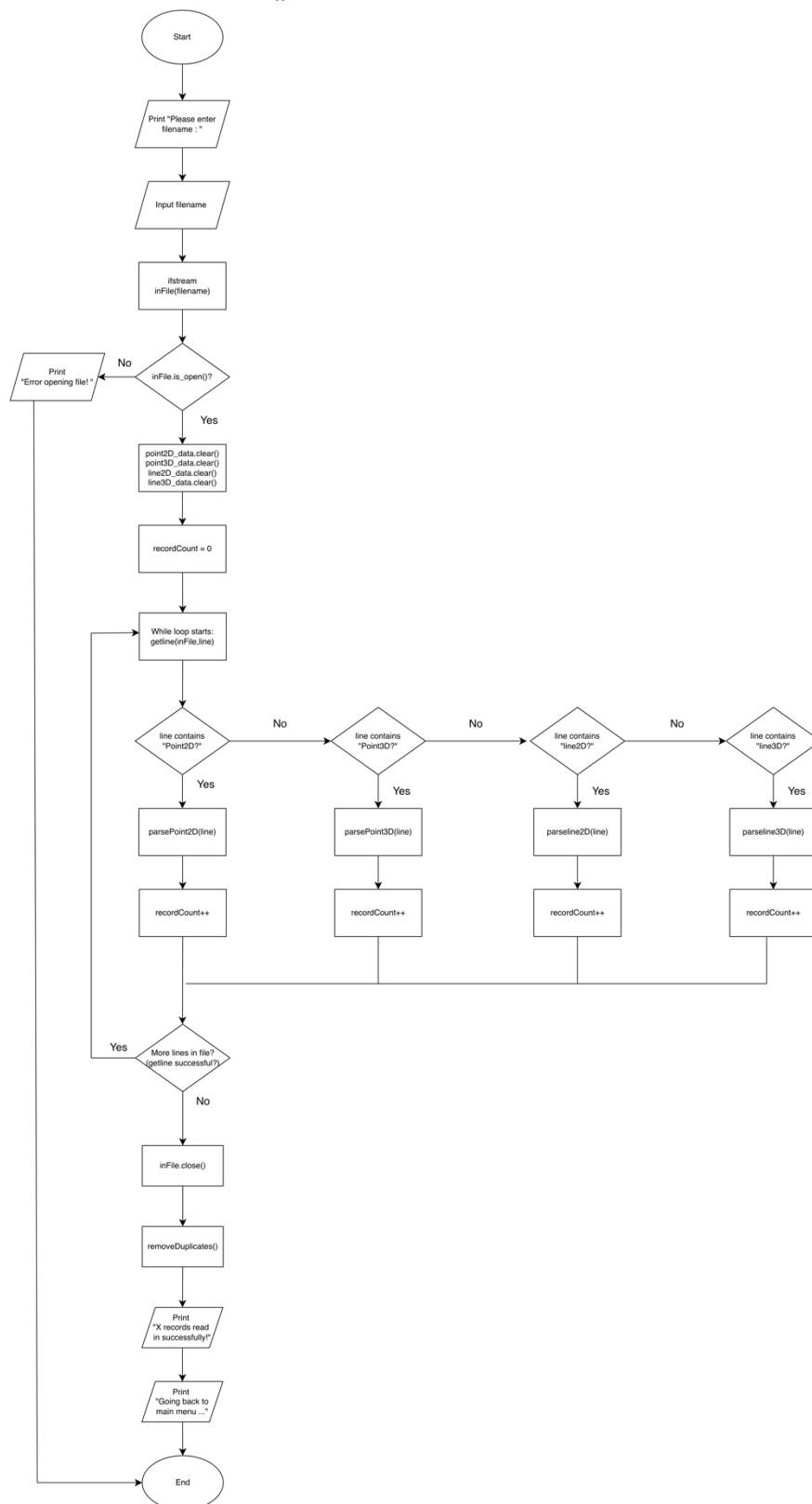
- `main()` - Program entry point → Loop continues while choice is 1-6 → Exits when choice < 1 or choice > 6

Part 2: Flowchart

Flowchart 1: main() Function



Flowchart 2: readData()



Flowchart 3: removeDuplicates()

